

Review of Accident Detection Methods Using Dashcam Videos for Autonomous Driving Vehicles

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Abstract—The need for a reliable system to detect high-risk incidents in complex settings like roadways, which are infrequent but potentially dangerous, has arisen due to the occurrence of rare hazardous events. This system would empower self-driving cars to function autonomously over extended periods without human involvement. Among these hazardous occurrences, accidents have received the least attention due to their rarity and diverse nature. Recently, dashboard cameras (dashcams) have gained recognition in academic circles as a cost-effective and accessible solution to enhance the safety of autonomous vehicles when handling accidents, since they are now commonly found in most vehicles. This review presents the progression of concepts in this domain, tracing its development from early ideas to cutting-edge techniques. It categorizes these approaches into supervised, self-supervised, and unsupervised learning. Furthermore, the review thoroughly examines evaluation criteria and available datasets, providing a comprehensive comparison of the strengths and limitations of different methods. Ultimately, the review proposes potential avenues for future research in this field.

Index Terms—Accident detection, autonomous vehicles, deep learning, traffic anomaly detection, dashcam videos.

I. INTRODUCTION

IN THE context of roads and transportation, human behaviour is crucial in ensuring road safety, but predicting it is complicated because it is influenced by interactions with other road users such as vehicles and pedestrians, as well as the surrounding environment. According to statistics in [1] and [2], a significant percentage of accidents are caused by human error. Therefore, it is essential to comprehend these interactions to facilitate the transition to Level 3+ of automation (classified by Society of Automotive Engineers (SAE) [3]), where autonomous vehicles (AV)s must safely control all driving functions.

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Autonomous vehicle systems rely on processing a significant amount of perception data to detect potentially high-risk situations, such as object detection, classification, and localization. However, this data can be incomplete, uncertain, or noisy, making it challenging to achieve early and robust detection of potential accidents. The complexity of the driving domain—including an infinite number of unusual scenarios and unseen context—adds to the difficulty of hazardous event detection of autonomous vehicles [4], [5], [6], [7]. Furthermore, as it is impractical to manually define all risky situations, a flexible framework is crucial for accommodating the ever-evolving driving environment.

A. Term Clarification: Difference Between Dashcam, Front Camera and Rear Camera

A dashboard camera (commonly referred to as a dashcam), a front camera and a rear camera are devices used for capturing video footage, but they serve different purposes and are typically used in different contexts with respect to their purpose, location, application and functionality.

From one perspective, dashcam is specifically designed to record video footage (such as road incidents, accidents, and the surrounding traffic) continuously from the perspective of a vehicle's dashboard or windshield in motion. In contrast, front camera is a general term that can refer to any camera positioned on the front of a device or object and is not limited to vehicles as it can be found on various devices such as smartphones, laptops and security systems. Therefore, front camera on a smartphone can be used for selfies and video calls, it is used for video conferencing on a laptop, and in security systems, it can capture video footage of an area for surveillance purposes. Considering the rear camera's perspective, it is primarily used to assist drivers when reversing or parking by providing a view of the area behind the vehicle to help the driver avoid obstacles and ensuring safety while backing up. These types of cameras are mainly mounted near the license plate or on the rear bumper and have a narrower field of view.

B. Why Using Dashcams for Accident Detection?

As the most risky situation for autonomous vehicles, the detection and anticipation of accidents is still a grey area in both industry and academics. Perhaps the biggest hurdle



Fig. 1. Dashcams are reasonably-priced, widely accessible, easy to install, occupying limited space and while they are not blocking driver's vision, yet they record the most important moments. (Source: thinkwarestore.com and pioneerelectronics.com).

in overcoming this issue is lack of sizeable real dataset of accidents providing information from all state-of-the-art sensors, including lidars due to high price of such sensors and rareness nature of accidents. Having said that, dashboard cameras (dashcams), in contrary to other sensors, have become widely available in recent years installed in almost every car (Figure 1). This unique advantage of their availability in cars has helped dashcams to be present recording many accidental scenes and their database has ever been growing. As deep learning models have always shown improvement by providing larger dataset and considering the chance of sensory fusion of dashcam dataset, there is no doubt that dashcam based models will become more reliable and overall decision-making of autonomous vehicles in accident scenarios will improve significantly.

C. Challenges and Promising Future of Dashcams

In recent years, scholars have been working towards tackling the problem of accidents by creating hazard-predicting techniques that rely on dashboard cameras [8]. While their study can be regarded as the fundamental framework for employing dashcams in accident detection and anticipation, the process of identifying an accident itself proves to be exceptionally demanding due to the dynamic background [9]. This dynamic background stands in stark contrast to surveillance cameras with static backgrounds, creating complexity in distinguishing between hazardous events and regular scenes. Given the outlined problem, the objective of this paper is to aggregate all pertinent research within the realm of traffic accident detection (TAD) through dashcams. This paper intends to construct a comprehensive evolutionary survey of TAD, with the goal of expediting research efforts aimed at surmounting present obstacles. The attainment of a dependable and robust method for dashcam-based TAD could substantially enhance the affordability of the overall autonomous vehicle system. Such an achievement would mark a significant advancement in the widespread manufacturing of autonomous vehicles.

D. Dashcam Models: From Hazardous Incident Detection to General Anomaly Detection With a Focus on Accidents

Typically, dashcam models employ a hybrid deep learning framework that merges convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to identify potential hazards in videos recorded by dashcams, serving as a shared foundation. These methods involve parsing images using a CNN to extract appearance features and localizing

objects. The extracted spatial features are subsequently fed into an RNN to capture temporal patterns and understand how these spatial features evolve over time. For instance, [10] used Mask R-CNN for semantic segmentation and then used a CNN-LSTM network to classify dangerous lane change detection based on the frame overlays with these masks. Additionally, [11] proposed a risk assessment method based on the impact of any risky object on drivers' behaviour. Using a cause-and-effect approach, the risk of identified objects is assessed with a two-step framework that determines the reaction of the driver (effect) based on the risky object (cause). However, unguided deep learning models can train spurious patterns [12], and CNN models may fail to extract unobservable contextual relations in their features, as shown by studies of [13] and [14]. In their recent study, [15] proposes that when it comes to detecting high-risk incidents, such as accidents or abnormalities, the primary challenges lie in addressing the issues of imbalanced distribution and diverse anomaly classes. As discussed in the above studies, the primary goal of hazardous incident detection is to identify and respond to events that pose a threat to safety or the environment. These incidents can include accidents, emergencies, or any situation that may lead to harm.

Looking from broader perspective, accident detection is a kind of anomaly detection in the context of dashcam videos that can be applied in various domains to recognize deviations from regular patterns. This involves identifying comparatively straightforward visual anomalies, such as abrupt movements in a specific area of the video frame or the presence of visual artifacts or objects that were not included in the training dataset. Contrary to that simple visual anomaly perspective, driving scenes are considered with continuous movement, and anomalies are frequently defined by intricate interactions among participants in traffic which show a significant imbalance between normal driving scenes and accident scenes. To tackle these challenges, various methods have been suggested, which can be categorized into three primary types: video-based, segment-based, and frame-based detection approaches.

1) *Video-Based Accident Detection*: In video-based approaches, the whole video is given as input for TAD. These methods are generally divided into conventional/classical classifier and reconstruction error-based models. Conventional classifiers operate under the assumption that all normal instances belong to a single category, while reconstruction error-based models rely on the premise that abnormal samples will exhibit higher reconstruction errors compared to normal samples. One-class classification, as described in prior works [16], [17], [18], refers to methods that construct a multidimensional representation for normal data, with the objective of making that representation as compact as possible. During the testing phase, any data points that fall outside the learned multidimensional representation are flagged as anomalous. The deficiency of such classifiers is their need to have features extracted separately [19]. To make an end-to-end video-based approach, some literature came up with reconstruction error-based models that extract a feature map from input video and distinguish an accident/abnormal video from a normal video based on their reconstruction

error (large error for abnormal video and small error for normal video). These methods can further be divided into Generative adversarial network (GAN) based, such as [20] and Autoencoder based networks, such as [21]. Having said that, in facing dashcam accident videos, which have imbalanced distribution and contain dynamic, complex backgrounds, video-based approaches cannot cater to the need of TAD.

2) *Segment-Based Accident Detection*: Looking at accident detection from segment-based categories such as in [23] and [24], these techniques were developed to tackle the challenge of imbalanced distribution in traffic accident detection using a weakly-supervised framework that incorporates Multiple Instance Learning (MIL) principles. In the segment-based TAD, each video is separated into clips where an anomalous video can have normal clips before and after the anomalous clip of that video. Using class labels of anomaly and normal for the whole video instead of each video segment/clip, in their research, authors aim to directly learn the anomaly score for each video segment, with the goal of reducing the time needed for manual annotation of each segment. To implement their idea, in their study [24], first video segments are fed to MIL, extracting segment features, and then a 3D convolutional network is applied to them. Each segment receives an anomaly score from which the anomaly segment is determined. Reference [23] developed this idea by introducing the optical flow of each segment as a segment feature and applying attention to the MIL method which improved the overall performance. A more intricate approach was proposed in [25], which involves a dual branch network consisting of a spatial-temporal dynamic subnetwork and an interconnected dynamic subnetwork, both of which are trained using MIL techniques. While the former network is fed with the video segments, the latter has the input of feature maps that represent the interaction of pedestrians and the surrounding environment. More completely, [26] introduced the location of the anomaly to segment-based literature. Although such segment-based methods can be useful in the detection and localization of TAD, they cannot deal with multiple anomalies in a complex scene.

3) *Frame-Based Accident Detection*: In frame-based approaches like [27] and [28], the current frame is constructed using the previous frames and the anomaly is detected as the discrepancy in reconstruction between the generated current frame and the original frame. While such frame-based approaches can accurately determine the spatial and temporal location of abnormalities, they are still time-consuming, same as video-based and segment-based methods. Frame-based methods same as video-based methods fall into two main categories: Classifiers and Reconstruction Error-based methods, where classifiers have been used in supervised learning [15], [29]. As the dominant category in the literature of frame-based approaches, reconstruction-based methods require image or frame generation in networks, these can be broadly classified into two subcategories: 1) **Autoencoder networks** [30], [31], [32], [33] and 2) **Generative adversarial networks (GANs)** [34]. Autoencoder networks take input frames and compress them into a condensed representation, referred to as a “bottleneck” or latent vector, which contains fewer

dimensions but retains crucial features. This latent vector is then used by the decoder to produce approximate future images or frames. Despite its usefulness, autoencoders are susceptible to overfitting and the vanishing gradient problem, which are common issues in neural networks [35].

On the other hand, GANs aim to generate synthetic data that closely resembles real data, achieved through a training process involving adversarial learning between the generator and discriminator. To address the issue of detecting anomalies, [36] proposed a framework for predicting frames that incorporate standard spatial constraints based on intensity and gradient, as well as a temporal constraint based on motion. Their criteria for detecting anomalies are based on the consistency of the optical flow between the predicted frames and the ground truth frames. Consequently, this integration of spatial and motion constraints helps distinguish anomalies from normal events as anomalous scenes cannot become predicted based on the learned pattern. Figure 2 shows a distinction between normal and anomalous scene. As discussed, even a frame prediction method is of high computational cost. Recently, state-of-the-art methods suggested object-based prediction [22] and later in more complex methods, researchers have introduced object and scene interaction features [37] and applied ensemble of methods [37], [38]. The summary of the whole TAD categorization is depicted in Figure 3.

E. Purpose and Contribution of This Survey

While some studies like [39] provide a broad picture of solving the problem of risky incident detection with graph-based methods, this study tries to provide a comprehensive evolution of literature that has led to state-of-the-art methods in TAD using dashcams. Furthermore, in addition to the mentioned categories of TAD—comprising video-based, segment-based, or frame-based—same as all deep learning subjects—according to the usage of dataset labels, methods can also be divided into supervised, unsupervised, self-supervised and other in-between approaches. Regardless of their approach within the label-based categories (Unsupervised, self-supervised, supervised, or any mixed versions), or content-based categories (video-based, segment-based or frame-based methods), as they will be discussed in details in this study, all TAD methods are formed upon object detection, object tracking and spatial-temporal features that are extracted from the whole frames or objects and their motion (Figure 4).

With the general overview of dashcams and accident detection provided in this section, the rest of this paper tries to discuss about strengths and weaknesses of available TAD methods using dashcams in the literature that led to four state-of-the-art methods: [22], [37], [38] all of which are frame-based, error reconstruction-based models with the GAN architecture, while the model of [15] is also considered frame-based, but classifier based method. To keep the consistency in this survey, the sections are provided according to chronicle of publications and each section discusses the evolution of methods on that category. As a result, the rest of this paper is organized as follows: Section II elaborate on Unsupervised



Fig. 2. This figure shows an example of how a normal scene turns into an anomalous scene. In the first sampled frame (from right) the truck is crossing the intersection from right to left with orange direction and the car is passing in the direction of yellow arrow (TAD evaluation curve shows low values). However, in the second sampled frame when accident has taken place although truck is still following the orange direction, the car not only has been deviated toward red curved arrow from its yellow direction but also has been deformed (highlighted with red oval). Such deviations from normal motions and deformations of objects' appearance make a scene anomalous, which is reflected in the TAD curve. As can be seen in third frame, the anomaly has intensified with the additional deviation of truck from its orange path and change of appearance in the frame with car's part being separated in the air (TAD curve shows even higher values). Adapted from [8] and [22].

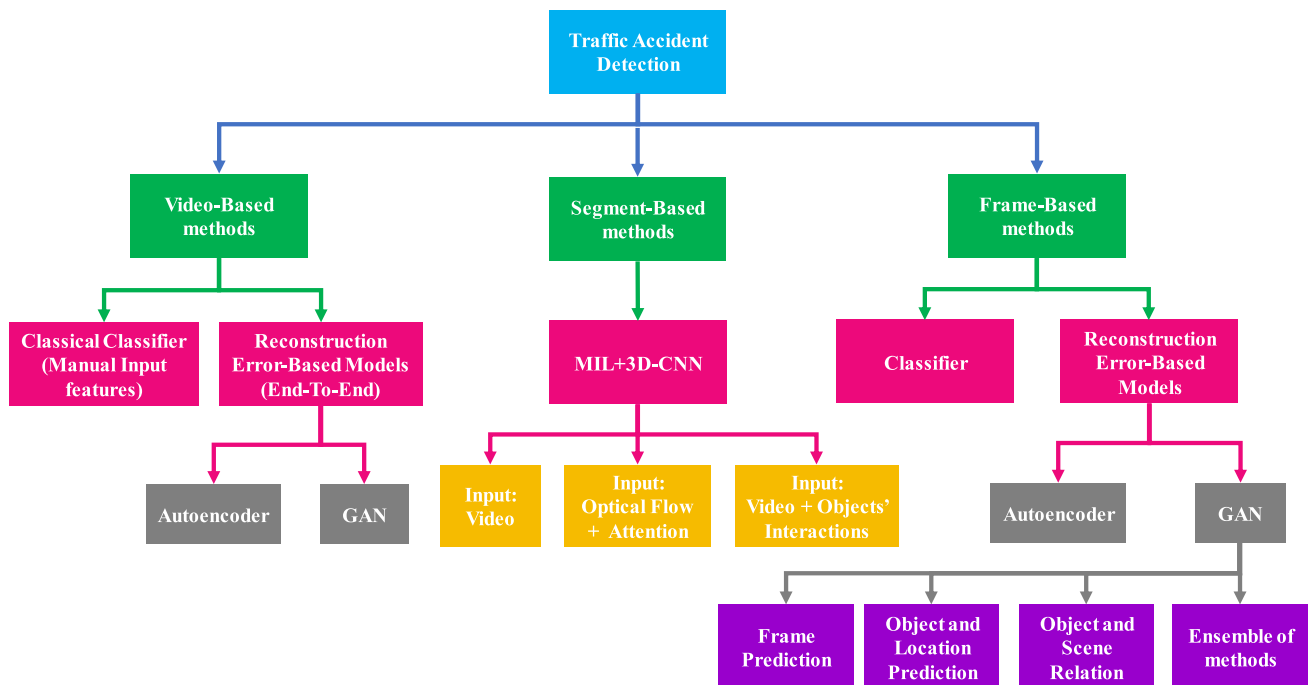


Fig. 3. Summary of Traffic Accident Detection Methods. (It should be noted that TAD methods under “Segment-based methods” category have specifically applied the method “MIL+3D-CNN” and they are differentiated based on the input of the method.)

video-based Traffic Accident Detection (TAD) methods [22], [38] and all relevant literature. Section III discusses about where Self-Supervised video-based methods stem from to deal with TAD [37], [40]. The supervised frame-based learning methods, [15], [29], are discussed in Section IV. Next, in the V an in-detail explanation of the available dataset is presented. Section VI provides the applied evaluation metrics in TAD.

Then comes the discussion and comparison of the state-of-the-art methods in VII, followed by discussions on future works in VIII. Finally, the conclusion of this study is delivered in IX. Overall, the contribution of this survey can be succinctly outlined in the following way:

- 1) This paper presents the evolution of all available dashcam-based TAD methods in the literature

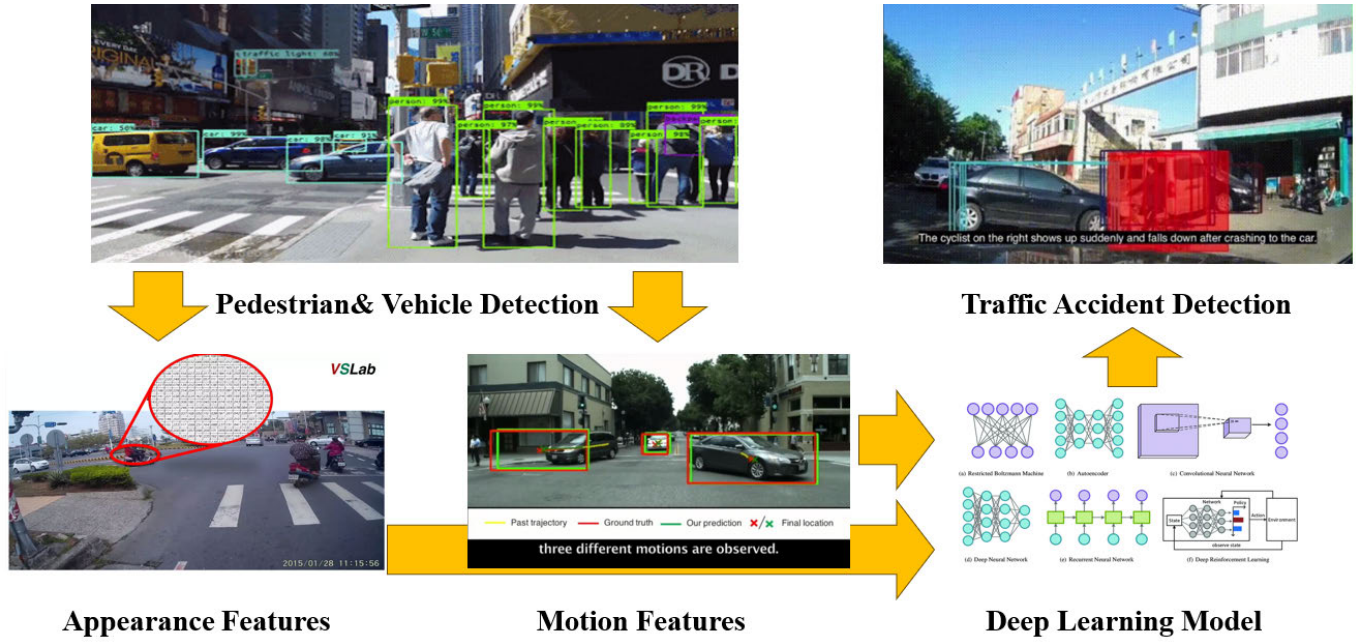


Fig. 4. The actual procedure of Traffic Accident Detection methods. Adapted from [8] and [22].

- 2) All catered and applied dataset in dashcam-based TAD methods whether publicly available or been of limited access are discussed and compared in detail.
- 3) Then comes the pros and cons of evaluation metrics in dashcam-based TAD methods.
- 4) Finally, the comparison, strengths and weaknesses of the state-of-the-art dashcam-based TAD methods and possible future directions are discussed, accordingly.

II. UNSUPERVISED TRAFFIC ACCIDENT DETECTION

From a general perspective, unsupervised machine learning methods are considered as those methods that during the training, the model is not given any output label. In the context of TAD, unsupervised methods are those whose dataset videos are not having the label “accident” or “non-accident/normal”. Going into more detail, some datasets have not just labelled videos, but they have a reach annotation per frame, such as DoTA [38] (There is a thorough explanation of different datasets in section V). The mindset behind unsupervised methods of TAD is that labelling all videos/frames can be so time-consuming and any method that relies on and is trained with labels would not be able to generalize well and would not be able to detect any new accident type. With this perspective, researchers considered the unsupervised TAD as a type of anomaly detection in which the model learns to be trained with non-accident events and recognize any abnormal event as an accident. However, anomaly detection has been mainly applied to surveillance camera datasets, which have a static background and applying those methods for accident detection with dashcam datasets which have dynamic background can be challenging. To deal with this challenge, researchers of [22] tried to generalize the idea of anomaly detection in videos based on future frame prediction which was initially suggested in [36]. In fact, the accident detection method of [22] is a combination of anomaly detection in [36]

based on future object localization in [9] and tracking objects method introduced in [41].

A. Fundamentals of Unsupervised TAD

The core structure of [22] relies on the foundational design of AnoPred [36]. AnoPred, in turn, takes advantage of both spatial and motion constraints. Within the AnoPred framework, the spatial constraint emerges through an encoder that captures both intensity (reflecting pixel likeness, i.e., $L_{Int} = norm_2(I_t, \hat{I}_t)$) and gradient features (preserving edges) from predicted and ground truth frames, i.e., $L_{Grad} \sum (pixel-wise\ gradient(I_t, \hat{I}_t))$. These features are then measured using the l_2 and l_1 distance metric, respectively. Regarding the motion constraint, the comparison between predicted and ground truth frames is facilitated by Flownet [42], i.e., $L_{Flow} = norm_1(Flownet(I_t, \hat{I}_t))$. This component extracts optical flow information from frames and is subject to penalization through the Least Square GAN module (L_{Disc} Eq. 1, L_{Gen} Eq. 2, PI is the set of patch indexes in image $\hat{I}_{(t+1)}$ and PS is Patch Score function that produces a scalar value for each patch). As a result, the final loss function of [36] can be considered the Eq. 1 as the discriminator loss and Eq. 3 as the total generator loss function. The ultimate comparison of their architectural performance is quantified using peak signal-to-noise ratio (PSNR) [43].

$$L_{Disc}(I_{t+1}, \hat{I}_{t+1}) = \sum_{(i,j) \in PI} \left\{ \frac{1}{2} [PS_{(i,j)}(\hat{I}_{(t+1)}) - 1]^2 + \frac{1}{2} [PS_{(i,j)}(\hat{I}_{(t+1)})]^2 \right\} \quad (1)$$

$$L_{Gen}(\hat{I}_{t+1}) = \sum_{(i,j) \in PI} \left\{ \frac{1}{2} [PS_{(i,j)}(\hat{I}_{(t+1)}) - 1]^2 \right\} \quad (2)$$

$$\begin{aligned}
& L_G(I_t, I_{t+1}, \hat{I}_{t+1}) \\
&= \sum_{(i,j) \in \{Int, Grad, Flow\}} \left\{ \lambda_i L_i(I_t, I_{t+1}, \hat{I}_{t+1}) + \lambda_{Gen} L_{Gen}(\hat{I}_{t+1}) \right\}
\end{aligned} \tag{3}$$

To make this backbone architecture (AnoPred) adapted with TAD application, authors of [22] conducted a research in [9] that predicts future location of vehicles in the scenes instead of predicting the whole frame, which is computationally more efficient. Delving further into the specifics, within the context of [9], a combination of dense optical flow (to capture motion-related information) and vehicle localization (to extract alterations in appearance) is employed. This fusion is utilized to predict the forthcoming positions of observed vehicles in dashcam videos. However, given that one of the major challenges within the dashcam dataset involves obscured scene participants, an object tracking strategy known as DeepSort [41] is also implemented. This technique aims to ensure consistent optical flow in the presence of occlusions. Indeed, in the TAD (Traffic Anomaly Detection) approach, not only [22] but also [37], [38], and [44] all incorporate DeepSort [41]. DeepSort, an extension of the Simple Online and Real-Time Tracking (SORT) method [45], addresses the issue of unstable identity tracking in SORT. SORT often encounters difficulties in maintaining consistent object identities due to frequent identity changes and resulting uncertainties during prolonged occlusions. In contrast, DeepSort [41] operates in two phases: an offline step, where each object's evaluation metric is computed, and an online phase, where these metrics guide object tracking. According to this method, if a tracked object is lost or obscured, a Kalman filter predicts potential object positions in subsequent frames within a limited time window (age) based on proximity to the lost object. If no plausible match is found, the object is considered to have left the scene, and its track is terminated. Moreover, when new object tracks emerge, they undergo evaluation against existing tracks. Only if deemed genuinely new are they assigned reliable metrics; otherwise, they are removed after three consecutive frames. In the context of [9], the application of the DeepSort [41] object tracking technique on bounding boxes leads to the encoding of depth and distance attributes of vehicles from the viewpoint. Simultaneously, by encoding dense optical flow, alterations in vehicles' dimensions, appearance, and motion at the per-pixel level are extracted. During decoding, temporal bounding boxes are predicted in relation to their previous frames, extrapolating their path. To account for the ego-vehicle's motion in predicting future bounding boxes, a frame-by-frame camera coordinate transformation is calculated.

B. Unsupervised TAD: The Baseline

Drawing inspiration from their object localization technique outlined in [9], the researchers in [22] introduced an unsupervised methodology for analyzing dashcam videos without the need for labeled data, named Future Object Localization based TAD (FOL-TAD). Their approach not only

accommodated the dynamic nature of dashcam video backgrounds, i.e., adapting surveillance camera anomaly detection methods from [36] to dashcams, but also transformed the problem of accident detection. They shifted from a one-class classification paradigm, which relies on meticulously labeled training datasets and confines anomaly identification to predefined categories, to a reconstruction error-based model. In essence, their method focused on predicting future object locations within the scene. If the anticipated positions of scene objects significantly deviated from their actual positions due to abrupt changes in object speed or position, the scene was considered anomalous—indicating a potential accident. To enhance the effectiveness of their approach in rapidly moving driving contexts, they restricted predictions to objects within bounding boxes ($X_t = [C_t^x, C_t^y, \omega_t, h_t]$, observed object's bounding box) rather than the entire frame, also forecasting their trajectories. Additionally, leveraging insights from their prior work in [9], the researchers integrated the movement of the camera within the ego-vehicle with the positioning of other objects in the scene. This integration was reflected in the anticipation of future bounding box positions for the current frame. Elaborating further, information from the bounding boxes in the current frame (I_t) and optical flow data, extracted through a region-of-interest pooling (RoIPool) operation, were separately encoded. In other words, observed image evidence O_t which is a relative motion vector that is expanded from each bounding box ($O_t = ROI Pool(I_t, X_t)$). The resulting hidden states were then input into a location decoder to forecast the future positions of the bounding boxes. With this localization technique established, the researchers aimed to identify accidents by detecting significant deviations in the trajectories and locations of objects from their ground truth frames. While their approach seemed theoretically sound, it encountered challenges in accurately identifying accidents due to the dynamic background introduced by the ego-car. Among the three evaluation metrics proposed in their approach, one particularly effective measure involved computing the maximum standard deviation (STD) across all predicted bounding boxes for each object within a frame, based on its previous frames. Taking an average of these STD values across all objects' bounding boxes provided a means to gauge the presence of anomalies or accidents in the scene. The summary of their method is as follows:

$$Trck[i].X_t = \begin{cases} \text{initialize,} & \text{if } i \in C \text{ and } i \notin D \\ X_t^{(i)}, & \text{if } i \in C \text{ and } i \in D \\ Trck[i].\hat{Y}_{(t-1)}, & \text{if } i \notin C \text{ and } i \in D \\ & \text{and } Trck[i].Age < A \end{cases} \tag{4}$$

$$Trck[i].\hat{Y}_t = FOL(Trck[i].X_t, O_t, E_t) \tag{5}$$

$$FOL - MaxSTD = \frac{1}{N} \sum_{i=1}^N \max_{\{C^x, C^y, \omega, h\}} \mathbf{STD}_{j=1}^{\delta}(\hat{Y}_{t-j}^{(i)}) \tag{6}$$

C. Unsupervised TAD: The State-of-the-Art

In a later examination by Yao et al. [38], the importance of pinpointing the anomaly's location within the frame or

identifying the involved road participants in practical driving assistance applications was emphasized. This was in contrast to earlier methods that utilized dashboard cameras [8], [46], [47] to identify anomalous frames. In [38], researchers aimed to not only detect anomalous events but also localize anomalies within frames or identify anomalous road participants. On one hand, prior unsupervised video anomaly detection (VAD) methods that operated on a frame-by-frame basis [36], [48], [49], [50], [51], [52] struggled to predict or reconstruct frames from rapidly moving dashcam videos. Conversely, object-centric frame-based approaches like the one presented in [22] heavily relied on object tracking techniques to determine object motion in the scene, which still encountered challenges with occlusions. In response, the researchers in [38] proposed a synthesis of their previous object-based method from [22], specifically the STD-based FOL, with the frame-based approach of [36], which is called FOL-Ensemble. Their ensemble-based FOL computed an average anomaly score by mapping the object-based anomaly score from [22] to a pixel-level anomaly score, similar to the approach in [36]. Furthermore, the study [38] introduced a comprehensive traffic VAD dataset called Detection of Traffic Anomaly (DoTA). This dataset included annotations for temporal, spatial, and categorical information, making it useful not only for anomaly detection but also for action recognition and online action detection (discussed in section V). Additionally, the researchers introduced a novel spatiotemporal evaluation metric for VAD, known as STAUC, which offered several advantages over traditional evaluation metrics (discussed in section VI). It is worth noting that although methods in unsupervised TAD are free of any accident/normal labels for frames or clips, the preprocessing step of object detection, which is not considered as part of TAD methods, still use bounding boxes' annotation for detecting objects.

III. TRAFFIC ACCIDENT DETECTION VIA SELF-SUPERVISED LEARNING

A. Unsupervised vs Self-Supervised TAD: The Difference

Since both unsupervised and self-supervised learning are techniques that do not rely on labeled data, this section first starts with providing a distinguishing explanation of these two methods. The fundamental contrast between self-supervised learning and unsupervised learning lies in the manner they generate learning signals. In self-supervised learning, the model generates its own signals for learning from the available data, whereas unsupervised learning focuses on identifying patterns or structure within data without explicit labels or guidance, such as exact location of objects, light intensity and exact color [53]. It is important to recognize that distinguishing between these two concepts can occasionally be nuanced, and the boundary between them might not always be precisely defined. Moreover, certain techniques could belong to both categories, as they utilize a blend of self-generated signals and unsupervised exploration of data. In the context of TAD, such feature labels are extracted from appearance and motion within objects' bounding boxes and also from the whole frame. What can make these features more reliable is to extract independent

ones by using similarity graph such as in [37] and using them in ensemble way [37]. Due to the lack of application of anomaly detection concepts in dashcam videos and even usage of unsupervised and self-supervised methods, studies of [54] suggested an unsupervised baseline to fill this gap. Inspired by their research, the initial idea of self-supervised learning in Traffic Accident Detection (TAD) was first introduced in [40] which was then evolved into [37].

B. Fundamentals of Self-Supervised TAD

As the only accident detection study on trucks' footage, the scholars from [54] introduced a comprehensive compilation of truck dashcam videos named RetroTrucks, which constitutes a curated dataset featuring various instances of both regular and accidental truck driving scenarios. They employed two techniques, namely one-class classification and reconstruction-based anomaly detection, using existing static-camera datasets as well as the RetroTrucks dataset. Utilizing the Support Vector Data Description (SVDD) technique proposed by [17] and the inflated 3D convnet (I3D) introduced by [55], they demonstrated that one-class classification struggles to generalize effectively across a diverse range of normal scenarios, as evidenced by conventional datasets like [50] and [56]. However, in the context of their reconstruction-based approach, they established an autoencoder based on the I3D architecture. An investigation into cases of failure unveiled that the reconstruction-based method outperforms the one-class classification method in identifying anomalies that showcase new visual characteristics. However, sole usage of such a reconstruction based method is not efficient enough to detect anomalies that involve complex object interactions. Consequently, in order to enhance the detection of such object-related anomalies within the scene, by drawing inspiration from [57], the researchers introduced an additional parallel feature extraction process. This process incorporates region proposal network (RPN) and RoIAlign modules [58], [59] to capture these interdependencies, forming a similarity graph. This graph evolves over time through the utilization of a graph convolutional network (GCN) module. The inclusion of this supplementary module yielded improvements in both the one-class classification and reconstruction-based methods, as demonstrated by their findings.

C. Self-Supervised TAD: The Baseline

Drawing inspiration from the research conducted by [36] and [34], the investigation presented in [40] introduced a self-supervised technique rooted in Generative Adversarial Networks (GANs), which combines frame and object prediction. In contrast to the approaches taken by [9] and [22], which exclusively rely on object optical flow, the work of [40] took a progressive stride by not only considering object movements based on optical flow (utilizing the FlowNet 2.0 network [60]), but also by predicting object positions using spatial attributes. These spatial features were derived through Mask-RCNN and indexed using DeepSORT. By harnessing the spatial characteristics of both objects and frames, the authors aimed to leverage abrupt changes in the background as a meaningful cue for

accident detection. In essence, the approach proposed by [40] integrates both the global background context across frames and the localized consistency in object motion and appearance. The ultimate goal was to identify accidents through deviations in frame predictions, object locations, and object attributes. The effectiveness of their method was evaluated on the Honda Egocentric View Intersection (HEV-I) dataset as well as the A3D dataset.

D. Self-Supervised TAD: The State-of-the-Art

Building upon their earlier research detailed in [40], the scholars in [37] extended their TAD model, called Self-Supervised Consistency (SSC), to encompass various triggers of anomalies, including anomaly in spatial change (frame prediction), motion patterns (object detection and trajectory), and overall scene context (spatiotemporal interactions between objects), which creates a multi-task model. More precisely, the frame prediction module consists of a pair of encoders (*enc*) designed to extract spatial features from frames ($Z_t = \text{enc}(I_1, \dots, I_t)$) and the video's optical flow tensor (utilizing the Flownet 2.0 [60], i.e., $m_{1:(t-1)} = \text{enc}(\text{Flownet2}(I_1, \dots, I_t))$). Subsequently, these two sets of features are input into a decoder (*dec*) to generate anticipated future frames ($\hat{I}_{(t+1)} = \text{dec}(m_t + Z_t)$ and $m_t = \text{ConvLSTM}(m_{1:(t-1)})$). In their module for predicting future object localization, the researchers in [37] adopted the principles of the multi-object tracker DeepSort [41]. They proposed an adaptation of this technique by implementing a reverse application to address instances of incomplete or inaccurate object detections. Subsequently, the states of the objects are extracted through the utilization of the ROI pooling operator (*ROIp*). Ultimately, a fully-connected decoder (*FC*) is employed to calculate the forthcoming positions of objects in the next frame, $\hat{Y}_{(t+1)}$, as follows:

$$\hat{Y}_{(t+1)}^{N_T} = FC \left[\text{enc}\{X_{1:t}^{1:N_T}\} + \text{enc}\{\text{ROIp}(m_t, (X_{1:t}^{1:N_T}))\} \right] \quad (7)$$

In which $X_{(1:t)}^{1:N_T}$ are bounding boxes of N_T objects in time series frames of 1 to t .

Furthermore, by recapping previous works, it was observed that frame prediction-based TAD techniques, such as the approach presented in [36], can be susceptible to rapid camera movements and significant shifts in illumination during normal driving, leading to imprecise predictions. Conversely, trajectory-based TAD methods, exemplified by [22], identify accidents through deviations in the variance of object trajectories. Thus, To handle scenarios where accident-involved entities enter the scene within a very limited timeframe, and rendering trajectory prediction is ineffective, researchers in [37] introduced a novel visual scene context (VSC) feature as another aspect to their multi-task model of TAD. To extract VSC features, first background of object bounding boxes are removed by Deformable Convolution Neural Network (D-CNN) [61]. Then, ROIALign [59] is applied to extract objects' features and the rearranged (transposed) of these features are called embedding features (R_t). Next, inspired from the study of [54], embedding features and similarity

matrix of these features (S_t) are fed as nodes to GCN to create the final VSC features (g_t) Eq. 8.

$$\begin{aligned} R_{(t+1)} &= \text{Rearrange} \left\{ \text{ROIALign}(\text{DCNN}(X_{(t+1)}^{1:N_t}, I_{(t+1)})) \right\} \\ g_{(t+1)} &= \text{GCN}(R_{(t+1)}, S_{(t+1)}) \end{aligned} \quad (8)$$

where $X_{(t+1)}^{1:N_t}$ and $I_{(t+1)}$ are predicted object location and predicted frame at time $(t+1)$. To acquire the most efficient VSC features, and inspired by [36], they adapted GAN method, for which similar loss functions to that of Eqs. 1-3 were applied.

Finally, accident detection in the method of [37] is fulfilled through an efficient fusion of these modules following by the smoothing filter of Savitzky-Golay filter [62].

IV. TRAFFIC ACCIDENT DETECTION WITH SUPERVISED LEARNING

In this section, the methods that addressed the problem of accident detection by using accident annotated dataset are discussed. While [29] tried to deal with this problem by calculating the movement parameters of overlapped vehicles, [15] tried to provide a real-time accident detection method. Although their results can be superior in comparison to unsupervised or self-supervised ones, such methods may not be the future direction of TAD due to the realistic fact that not all videos with different types of accidents can be labeled. That being said, the techniques that have been used within each method can cite some ideas to be applied to any unsupervised/self-supervised methods.

A. Basic Supervised TAD

Establishing their supervised method on motion characteristics, researchers in [29] developed their accident detection method on acceleration, trajectory and angular variation of cars, named Movement (motion) Parameters method. To attain these key variables, they employ Mask R-CNN [59] for vehicle detection and employ the centroid tracking algorithm [63] to establish vehicle trajectories. According to their methodology, for an accident to occur, the bounding boxes of vehicles should exhibit overlap and align within the same lane, which they accomplish through the application of Aggarwal's lane detection technique [64]. Subsequently, they compute the Euclidean distance between centroids of vehicles in consecutive frames, which yields the trajectory parameter. This parameter is then used to determine the angle of trajectories between intersecting vehicles, indicating an unusual angle. Moreover, by utilizing the Lucas-Kanade method [65], they compute the pixel-based speed of vehicles within the scene, consequently deriving the absolute speed and change of acceleration for each vehicle. This information aids in identifying any abnormal change in acceleration. Their approach also detects anomalous trajectories by considering the temporal angle of a vehicle's trajectory. Ultimately, an accident occurrence is determined through a weighted function that combines these three aforementioned parameters.

B. Supervised TAD: The State-of-the-Art

As one of the state-of-the-art methods in TAD, the study of [15] provides a real-time method that suits Vehicular Ad hoc Networks (VANET) environment called Coarse-to-Fine clustering. Their frame-based supervised method first conducts coarse detection to find all probable accident frames, and then, determine the actual accident frames through the fine detection stage. In their coarse detection stage, [15] extracts motion features using Histogram of Optical Flow (HOF) features [27] using 15 consecutive frames and form a temporal ordinal features of frames ($V = \{v_1, v_2, \dots, v_m\} = HOF(I_1, I_2, \dots, I_T), m = T - 15$ and T refer to Total frames) along with their labels ($L = \{l_1, l_2, \dots, l_m\}$). To prepare these temporal features for later clustering, they applied a two-stage subnetwork neural node (SNN) structure as a feature representation, called refinement SNN(R-SNN)([66]). In its first stage (preprocessing), feature vectors of different feature extraction methods are normalized and aligned consecutively into a super-state vector. Next, in the second stage (Width-growth), first an R-SNN is initialized, and then additional R-SNNs are concatenated, iteratively. The origin of SNN technique has initially emerged from Network-in-network (NIN) concept of [67] in which every single node in a hidden layer is replaced by a neural network itself, called the subnetwork. This technique tries to alleviate the problem of over-fitting and getting stuck in local minimums of Deep Neural Networks (DNNs) models. This problem especially occurs when DNN models try to process a small-scale dataset which are dependent on the backpropagation (BP) learning. Leveraging the utility of R-SNN as a feature representation method [66], the researchers in [15] compress the temporal attributes into a more streamlined feature matrix. This matrix ($A_C^N = R - SNN\{V, L\}$), encompassing frame features and labels, undergoes clustering through a graph-based approach named self-representation constrained low-rank representation (SRLRR) [64], [68]. Subsequently, by employing the normalized cut (Ncut) method [69], potential frames (S_{Coarse}) with accidents are identified Eq. (9).

$$S_{Coarse} = Ncut \left\{ affinity[SRLRR(A_C^N)] \right\} \quad (9)$$

Moving on to the fine detection phase, the process unfolds as follows: Initially, Faster-RCNN (FRCNN) [58] is applied to these potential frames to identify objects within each frame. This involves utilizing both the log-likelihood of object detection and the regression loss pertaining to their positions in the frame. Following that, in conjunction with the established objects in the frames, the feature vectors extracted from the final convolutional layer of Faster-RCNN are sorted in two ways: one based on their spatial placement in the frame and the other according to the sizes of their bounding boxes, i.e., $\{F_{i-sp}^{obj}, F_{i-sz}^{obj}\} = FRCNN(s_i \in S_{Coarse})$ and $obj \in \{1 : N_T\}$ where N_T is the total detected objected in one of Coarse Frames. These grouped feature vectors, accompanied by their respective bounding boxes (X_i^{obj}), are individually fed into the R-SNN framework introduced in [66], resulting in the generation of efficient encoded features presented as a matrix, (i.e., $A_i^1 = R_SNN(F_{i-sp}^{obj}, X_i^{obj})$ and $A_i^2 = R_SNN(F_{i-sz}^{obj}, X_i^{obj})$).

Finally, the two resultant matrices are merged (*Concat*) into a comprehensive feature representation. With the aid of accident/non-accident labels provided (l_i) in the DoTA dataset [15], a supervised classification is executed using the Support Vector Machine (SVM) algorithm Eq. (10).

$$S_{Fine} = SVM_{i \in Coarse} [Concat(A_i^1, A_i^2), l_i] \quad (10)$$

V. DATASET

In this section, the evolution of dataset that has been introduced in the TAD literature is discussed. It is worth noting that since all accident detection and accident anticipation methods are mainly applying frame-based techniques, many of them provide the number of frames instead of length of videos as the information of each dataset video. In the upcoming discussion, we will explore how Figure 5 illustrates the annotation method used in the majority of publicly accessible datasets. In a general context, as it is shown in the figure 5, the difference between datasets comes from the length of their trimmed clips, their before and within and after accident intervals and most importantly their approach toward annotation. While in some datasets have only annotated accident objects from beginning to the end of clip, others have done the opposite by just annotating frames as normal/accidental and some have the annotation of both frames and objects. The amount of information that has been provided in each annotation also varies. In the following each of the available TAD dashcam dataset in the literature will be discussed in details.

A. SA

As the baseline of accident anticipation and inspiring for accident detection, [8] introduced the very first dashcam dataset, namely Street Accidents (SA)– also known as Dashcam Accident Dataset (DAD). They collected a variety type of accidents from Youtube recorded in six major cities in Taiwan with quality of (1280 × 720 p in resolution). In their dataset, the temporal annotation in each video frame comprises type of objects, bounding box of objects and flag them as accidental/non-accidental object through the course of time. More specifically, their annotation determines accidental objects from beginning to the end of video (not the beginning and the end of accidental frames). With overall number of 678 accident videos, they used less than 10% of videos for training object detector. As a result, from the rest (620 videos), 1,750 clips were sampled (35% accident positive), where each clip consists of 100 frames (4 seconds, i.e., 25 frame per second (fps)), and in those containing accident, clips were trimmed to have accident at last 10 frames. Their purpose on the clip trimming approach was to apply prior frames for accident anticipation and they further randomly divided these clips into a quarter for testing and the rest for training.

B. CCD

In another accident anticipation approach, Studies of [70] collected more accidental dashcam videos (in comparison to SA) from YouTube (1280 × 720 p in resolution) and is named

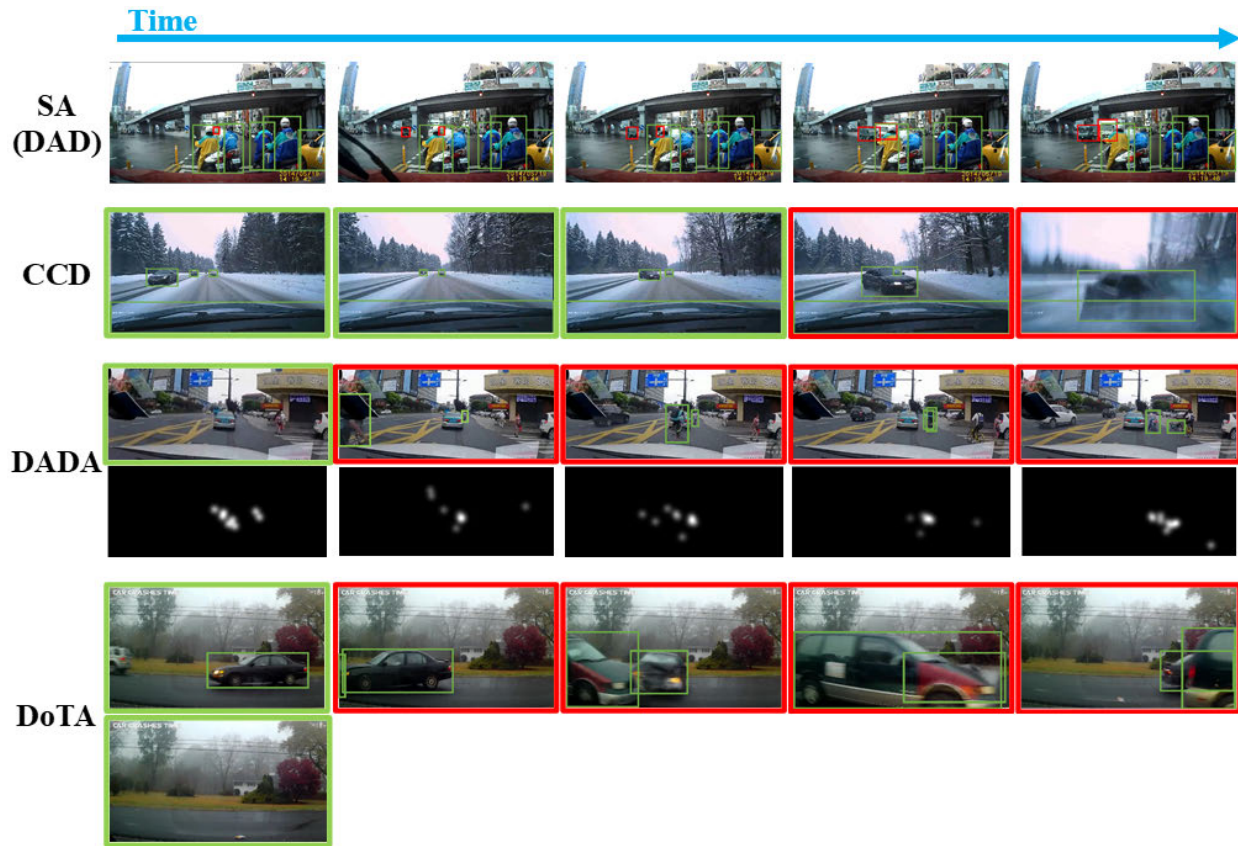


Fig. 5. First row: In SA (DAD) [8], accidental objects are flagged from beginning to the end of clip regardless of accident time, where red boxes are accidental objects and green boxes are uninvolved objects in the scene (annotation does not include accidental frames). Second row: In CCD [70], only accidental frames are annotated (not the accidental objects), where red boundaries are accidental frames and green boundaries are considered as normal frames. Third row: In DADA [71], frames are annotated in the intervals of before crash-object appearance (normal scene), before accident (crash-objects in the scene), accident interval and after accident frames (DADA has not released bounding boxes' information. N.B., their annotation considers frames as accidental from the appearance of collision objects in the scene, which are determined with red boundaries, while frames of before crash-object appearance and after accident are colored with green as normal frames). Row fourth: In DoTA [38], frames are divided into before accident (labeled as normal (green border), accident interval (red border) and after accident (green border). Based on annotator discretion frames are labeled accidental when accident is foreseeable, and bound boxes for non-accidental objects are not provided.

Car Crash Dataset (CCD). Their temporal annotation has a binary label, i.e., accidental frames are labeled 1 during the time interval of collision. Additionally, their annotation comprises the starting point of the trimmed clip in the original video, Youtube ID of that original video, time of recording (day/night), weather condition of footage and the involvement of ego-vehicle. It is worth noting that on their released version of annotation, [70] did not provide bounding box information of objects in video frames. Besides, in contrary to SA [8], CCD's annotation only determines accidental frames (not the accidental objects in the scene), and based on this released annotation, there is no information about objects in the scene. In their trimmed videos with 50 frames (5 second length, 10 fps), the occurrence of accident is randomly placed in last 2 seconds, generating overall 1,500 accidental video clips, which is half of the number of normal videos and the normal videos themselves are catered from BDD100K dataset [72]. Overall, the CCD comprises 4,500 video clips in which the ratio of testing videos is a quarter of that of training ones.

C. DADA

As another available dashcam dataset, authors of [71] introduced their dataset named DADA, consisting 2,000 videos

(14-second length) with the resolution of 1456×660 (30 fps). According to the purpose of their research, [71] wanted to add additional information to dashcam dataset using driver's attention. Therefore, different from datasets like SA [8] AND CCD [70] that were trimming frames to assign accident interval within last ten frames for accident anticipation, videos catered by [71] contain before and after accident intervals along with the accidental interval. In contrast with [8] and [70], in addition to objects' bounding boxes in each video frame which are not released in their available dataset), they created a visualized map of crash-object locations. The term "crash-object" in their study is defined as object involved in accident, and the accident interval in their study is started when half of crash-object appears in the scene and is ended when anomaly in the scene is over (normal movements of objects). According to their definition, in accidental clips, frames are flagged as accidental from the point that objects involve in collision appear in the scene. Moreover, their spatial object annotation does not include non-accidental objects in the scene. The DADA is categorized in 54 type of videos based on accident participants and they were further classified into two large sets, ego-car involved and ego-car uninvolved. Later, authors of [71] released their whole dataset in [47], where they called it DADA-2000.

D. CST-S3D

Recently, based on the study of [73], the DADA-2000, [71] and [47], has four commonly occurring mismatch annotations, which consists of labeling uninvolved ego-car as involved one, wrong category annotation, providing video games instead of real accident clips, and annotating accident video as normal. Their assessment also suggests that DADA-2000 contains only 520 videos of ego-car involved and 493 videos of ego-car uninvolved (overall of 1,013 annotated videos). With respect to shortages in DADA-2000 and owing to the insight of [73] where they suggests that safety of a vehicle stems from preventing ego-car involved incidents, [73] only re-annotates those videos of DADA-2000 where ego-car was involved. To strengthen the safety of their model, they also consider near-miss incidents as accident in their re-annotation. Furthermore, they suggested that there is no clarity to define before accident interval for ego-car involved accidents. As a result, in their re-annotation, they merged the whole time interval of before, within and after accident intervals of DADA-2000 into an incident interval and considered a two-second interval transient period, before considering a normal time interval.

E. A3D

Next, in [22], AnAn Accident Detection (A3D) was introduced as the very first dashcam dataset that was introduced for TAD and was collected (with 10 frames per second) with similar number of dashcam accident videos from Youtube to that of CCD [70] and with different number of frames in each clip. In opposite to CCD that determines accidental frames in each clip according to the interval that the collision has happened, in A3D of [22], the start point of accident interval is considered when accident is inevitable (which depends on three human annotators' consensus), not when the collision takes place, and the end point is defined (same as DADA [47], [71]) when scene is no longer anomalous (normal movement of objects). A3D annotation also includes start and end times of clip from original video, original video name, labels for accidental and normal frames, whether the ego-vehicle is involved, and whether ego-vehicle is the only vehicle involved in accident. Notably, A3D does not have any bounding box information of vehicles. Besides, although videos have been collected in different weather conditions, different road environment (urban/countryside) and with different road participants, (unlike CCD [70], SA [8]) the dataset's annotation does not provide any information in this regard. Following the instruction suggested in their github showed failure to download A3D due to unavailable Youtube links.

F. HEV-I

Before introducing their FOL-TAD in [22], researchers of that study focused on considering ego-vehicle's motion on predicting future location and dimensions of vehicles in the scene [9]. In their study, a non-accidental dataset was introduced, called Honda Egocentric View-Intersection (HEV-I). Recorded with the rate of 10 fps in San Francisco, U.S., HEV-I consists of 230 videos with 1280×640 p resolution of different lengths (10)-60 seconds), and is catered for distinguishing

vehicles in the scene with a continuous dynamic background. Due to fast-paced vehicles in the scene and limited point of view, [9] divided videos into two-second clips in which the location and dimensions of viewed vehicles in the second half are calculated according to assessment in first half using their trajectories.

G. DoTA

To address the issue of Youtube video URL lost and provide a richer dashcam video dataset, researchers of [22] introduced a new dataset called Detection of Traffic Anomaly (DoTA) in [38] which is an extension to their previous dataset (A3D). In comparison with older dataset, DoTA not only consists of more than 3 times the number of videos with previous mentioned annotation information (A3D), but also provides start and end of anomaly frames with labeling all frames (binary and with name tag), number of frames, day/night label, determining accident type (with considering ego-vehicle involved in all categories), bounding boxes of objects and optical flow of the objects in each clip. Same as A3D [22], in DoTA [38] no annotation is provided for different weather conditions, though dataset provides variety of the weather. All videos have been provided with 10 fps due to avoid repetition of information in close frames (even though original videos were 30 fps). Same as DADA [71], DoTA [38], based on their provided spatio-temporal annotation, has separated the time intervals of videos into before accident, accident and after accident, although they provided different preference on the beginning of accident. More precisely in DoTA [38] frames are considered accidental/anomalous when accident is inevitable and ends when anomalous objects become stationary in the scene or have left the scene. While they claimed such a start annotation is for the purpose of early accident anticipation, the term "inevitable" makes the decision on the start of anomaly arbitrary. N.B., only accidental objects are annotated.

H. CTA

Inspired by vehicles' history report of accidents catered by police department in [74], researchers in [75] created a video dataset called Causality in Traffic Accident (CTA) to analyze this purpose. Collecting almost 2,000 accident videos from Youtube (mainly dashcam footage and few surveillance cameras), [75] determined shot boundaries using FFmpeg (with manual justification) to label causal and effect frames. More precisely, in CTA annotation, an effect's start time is when a vehicle first experiences physical damage, and its end time is when there is no longer any activity related to the event happening to the objects involved. On the other hand, frames are labeled causal when any road participant shows anomalous behaviour and it ends at frame when scene becomes normal, though in actual cases cause's end frame is considered as beginning of effect frames.

I. NIDB

Instead of collecting accident footage, researchers in [76] gathered over 6,000 near-miss traffic incident database (NIDB)

from more than 100 taxis using a dashcam recording system that is triggered to record for 15 seconds when speed reduction rate, in an abrupt braking, exceeds half of gravity force equivalency. With overall of seven categories, they divided incidents into high/low pedestrian, high/low bicycle, high/low vehicle and normal clips, where normal implies none of the mentioned incidents has happened. According to their approach, high incident is when probability of accident was high and it required serious reaction of driver, while for low incidents, the scenario is opposite and system is required to give just an alarm.

J. STAG

In order to evaluate and refine their Spatio-Temporal Action Graph (STAG) approach, researchers in [46] created the Collision dataset. With average duration of 40 seconds, this dataset comprises ego-vehicle's accidents (about 750 videos) and near-miss incident (60 videos) that requires evasive action of the driver to prevent a crash. The Collision dataset was collected using dashcam which was triggered with Inertial Measuring Unit (IMU) and gyroscope readings, though distinguishing of accident and near-miss records were fulfilled by human annotator. To avoid duplication and examine few shot methods, they also applied a 5 fps filter to provide a low-rate dataset too. regardless of frame rate, each accidental or near-miss video was divided into three equal segments, with only one segment of accidental videos being positive.

K. RetroTruck

In contrary to all mentioned dashcam dataset in literature, [54] collected more than 470 dashcam footage (25 fps) of trucks with different duration (7 seconds to 2 minutes) from YouTube, called RetroTruck. Overall, all anomalous videos include ego-vehicle—comprising collision, near-misses, lane crossing incidents, rollover crashes— and they constitute less than 50% of dataset, where only 25% of these anomalous videos have temporal binary label (normal/anomalous) for the purpose of testing. Notably, although dataset is gathered from different weather and time of day, their annotation does not provide any information in this regard.

VI. EVALUATION METRICS

As discussed in the Future Frame Prediction II, to distinguish an anomaly from normal event in a video, the first step is to calculate the similarity of predicted frame and corresponding ground truth frame using PSNR. Then, all PSNR values of consecutive frames in a video are mapped to range [0, 1] (Normalization [43]), which are called anomaly score. All TAD methods use this anomaly score to provide their evaluation metrics.

A. Qualitative Analysis

As the research of [47], [71] have been conducted in studying the driver's attention in predicting and detection accidents, the qualitative analysis has always been part of all VAD studies to give an intuitive understanding of each method and how their approach tries to detect accident within time frame.

B. Area Under ROC Curve (AUC)

Being one of the most popular evaluation metrics in the anomaly detection literature, [48], [50], [77], [78], in the concept of VAD, after calculation of the anomaly score in each frame of a video, the temporal concatenation of these scores form the Receiver Operation Characteristic (ROC) curve. Area Under Curve (AUC), in this context, determines the location of possible accident through time interval. The higher the value of AUC, the better suggested model distinguishes accident from non-accident/normal events.

C. Accident Detection Rate and False Alarm Rate

As a way of assessing the performance of VAD, [29] suggested to use Accident Detection Rate (ADR) which is the ratio of truly detected accidents to the total number of accidents, i.e., ADR is actually parameter named Recall in the literature or TPR. Additionally, they applied False Alarm Rate (FAR) for comparison, which determines the ratio that system gives false alarm in comparison to total alarms that have been sounded. FAR is precisely FPR.

D. Time Metrics

While all the state-of-the-art methods use the AUC as one of their evaluation metric, the Coarse-to-Fine clustering approach [15] also suggested the comparison of methods with regards to Number of Parameters, Training Time and Average Detection Time, as a way to show compliance of different approaches to VANET environment.

E. Success Rate (SR) Curve

In addition to AUC, inspired by object tracking methods such as [79], the study of [37] also suggested to compare accident detection methods' performance using Success rate curve, which relies on bounding box. The overlap score, denoted by $S = \frac{r_t \cap r_a}{r_t \cup r_a}$, is defined as the ratio of the number of pixels in the intersection of the tracked bounding box and the ground truth bounding box to the number of pixels in their union. The intersection and union are represented by $\cap$ and $\cup$, respectively, and $|\cdot|$ denotes the cardinality or number of pixels in a region. To evaluate the performance of tracking algorithms, the number of frames with overlap greater than the specified threshold t_0 is counted over a sequence of frames. The success plot displays the success ratio of frames at different thresholds ranging from 0 to 1. Assessing the tracker's performance based on a single success rate value at a particular threshold, such as $t_0 = 0.5$, may not be a fair complete representation. Therefore, the area under the curve (AUC) of each success plot is utilized to rank the tracking algorithms. This metric provides a comprehensive comparison of the method's adaptability across various driving scenarios.

F. Average Precision (AP) and Mean Average Precision (mAP)

To make their evaluation more complete, [37] also applied Average Precision (AP) [80]. The AP (average precision) is a measure of how accurately a detection model identifies

TABLE I
SUMMARY OF COMPARISON BETWEEN ALL DASHCAM VIDEO DATASETS

Dataset	Availability	No. of Anomaly Videos	No. Frames	Annotations	Location (Source)	Website	Year
Street Accidents (SA) or Dashcam Accident Dataset (DAD) [8]	Public	620	62,000 (20 fps)	Temporal	Taiwan (Youtube)	SA Link	2016
Car Crash Dataset (CCD) [70]	Public	1500	75,000 (10 fps)	Temporal spatial (tracklets) causation	- (Youtube)	CCD Link	2020
Driver Attention in Driving Accident scenarios (DADA) [71]	Public	2,000 ¹	648,476 (30 fps)	Temporal spatial (eye-gaze)	- (Youtube, Youku, Bilibili, iQiyi)	DADA Link	2019
CST-S3D (Re-annotation of DADA) [73]	Public	39	1,170 (30 fps)	Temporal ego-car involved	- (Youtube, Youku, Bilibili, iQiyi, Tencent)	CST-S3D Link	2019
AnAn Accident Detection (A3D) [22]	Public	1,500	128,175 (10 fps)	Temporal	East Asia (Youtube)	A3D Link	2019
Honda Egocentric View Intersection (HEV-I) [9]	Private	230	79,991 (10 fps)	All Normal (No accident) urban intersections	US (San Francisco Bay Area)	HEVI-I Link	2019
Detection of Traffic Anomaly (DoTA) [38]	Private	4,677	731,932 (10 fps)	Temporal spatial(tracklets) categories	- (Youtube)	DoTA Link	2020
Causality in Traffic Accident (CTA) [75]	Public	1,935	731,932 (10 fps)	Cause & Effect	- (Youtube)	CTA Link	2020
Near-miss Incident Database (NIDB) [76]	Private	6,200	1.3M (10 fps)	Temporal categories	- (Driving Recorder in Taxis)	NIDB Link	2006-2015
Collision [46]	Private	803	4,015 (5 fps)	Temporal spatial (tracklets) categories	Large Cities in US (Dashacam, IMU and Gyroscope)	Collision Link	2019
RetroTrucks [54]	Public	474	11,850 (25 fps)	Temporal	- (Youtube)	RetroTrucks Link	2020

¹ In the work [73], authors claim there are only 1013 distinct samples in the DADA dataset.

accidents. It is calculated by averaging the precision scores of all testing videos. Precision is the ratio of correctly identified accident frames to all detected accident frames, which include both true positives and false positives. The detection method determines the positive and negative labels of detected frames using a pre-defined threshold. In this study, frames with an occurrence degree of traffic accidents greater than 0.5 are labeled as positive, and those with lower degree are labeled as negative.

G. Anomaly-to-Accident (A2A)

based on their temporal annotation of dataset, study of [37] introduced a metric that can be used for future accident anticipation studies. According to their definition, a temporal window called anomaly-to-accident (A2A) for each video sequence is started when the accident participant appears in the scene and as a result A2A can help predict accident by detecting anomalous object or behaviour before collision.

H. Spatial-Temporal Area Under Curve (STAUC) Metric

Being critique of efficacy of AUC, the study of [38] suggested a novel metric called Spatial-Temporal Area Under

Curve (STAUC). By their account [38], due to being averaged score in each frame, AUC dismisses providing any information/ accuracy of accident/anomaly region localization on the spatial axes, which is a fundamental need to assess the performance of VAD. By comparing the corresponding score map of true positive frames by applying different VAD methods, researchers of [38] showed that although AUC of different methods are similar, their corresponding maps are different, which advises that these methods' performance are considered the same from AUC point of view. This cause, brought them to introduce new metric, called STAUC. Inspired by the MVTec Anomaly Detection (MVTec AD) method of [81], The initial proposal was to compute a scalar known as the True Anomalous Region Rate (TARR), which is defined as the fraction of the anomaly score that corresponds to the union of all annotated bounding boxes relative to all the pixels in the frame at each time frame. This scalar was intended to indicate the proportion of the anomaly score that is present within the actual anomalous region. Initially, TARR was formulated for anomaly score maps that covered an entire frame. However, since object-based video anomaly detection (VAD) methods compute anomalies for individual objects, the concept of

TARR was extended to incorporate pseudo-anomaly score maps. This was achieved by computing the anomaly score of each pixel as the sum of a 2D Gaussian distribution of scores that were centered at the pixel's bounding box. Next, they defined spatio-temporal true positive rate (STTPR) which is a ratio of TARR of all true positive predictions to all ground truth positive frames. In fact, STTPR is a weighted TARR. From the values of STTPR and false positive rate the plot of ROC curve is formed which is called Spatial-Temporal ROC or STROC and the area under this curve is called STAUC. As the final step and to make their new metric robust against anomalous region size, they suggested to use an adaptive $N\%$ top anomalous pixels ($N_{adaptive}\%$) to be used in TARR computation. This means that for each frame, N is calculated as the proportion of number of anomalous region pixels to total number of pixels. The new metric $STAUC \leq AUC$ in the case that $TARR_t = 1 \forall t$. The summary of all evaluation metrics has been provided in Table II.

VII. DISCUSSION AND COMPARISON

Like every other computer vision problem, as presented in previous sections, Traffic Accident Detection (TAD) has been dealt with supervised, unsupervised, and as a newly introduced approach, self-supervised learning. Regardless of the approach in each method, all TAD algorithms start with an object detection method, followed by object tracking technique and, relying on them, some features are extracted to train a deep learning model to detect an accident as an anomaly.

A. Comparison of Largest Dataset: DADA and DoTA

Looking into the fundamental details of all mentioned algorithms, almost every study suggested their own catered dataset, each of which has been collected and annotated from mainstream video websites, especially Youtube, and based on their approach, being annotated accordingly. As a result, any further study in this field is highly dependant on the way annotation of dataset has been fulfilled, such as recent attempt of [73] where they reported wrong annotation of DADA, [71], and re-annotate part of it for their own research interest. This means that there is a lack of comprehensive consensus and protocol in the Intelligent Transportation Systems Society (ITSS) on TAD dataset annotation which will align all studies toward complimenting previous ones. (regarding the accuracy of annotation) and informative dataset with regards to its temporal, spatial, and categorical annotations of accidents for frames and objects in the videos, captured from different countries under different weather and lighting conditions, appears to be DoTA [38].

B. Dealing With Unreliable Object Detection and Tracking

From the object detection point of view, except the study of [15] which used Faster-RCNN [58], all other methods [22], [29], [37], [38], [40], [44] used Mask R-CNN [59]. This justify the high efficacy of this object detection method. As for the object tracking, while supervised movement parameters approach [29] uses centroid tracking [63], all the methods of [22], [37], [38], [40], and [44] have applied DeepSort [41]

technique (This technique is not utilized in Coarse-to-Fine method [15]). To alleviate the impact of TAD on unreliable object detection and tracking situation, researchers of unsupervised approaches tried to apply statistical operators, FOL-AvgSTD and FOL-MaxSTD variations in [22] and [44] and the FOL-Ensemble in [38] and [44], when their initial method faced mis-label objects due to losing some normal objects in the scene. On the other hand, to deal with unreliable object detection and tracking methods, researchers in self-supervised method of [37] took further steps in comparison to the unsupervised methods by using the DeepSort method [41] in backward, extracting visual scene context feature using GCN, make another ensemble method called collaborative Multi-Task consistency learning, and finally smoothing the visual scene context scores using a Savitzky-Golay filter [62]. Despite all the measures that the study of [37] took in their approach, they still struggled with background recognition in region of accident's future frame prediction. Consequently, their method fails in case of too large or too small scale objects to be discovered. Their results also down perform in low intensity environments (nights). This means that the additional extracted features in [37] has not actually helped with distinguishing accidental objects from dynamic background, in comparison of [22].

C. Supervised vs Unsupervised Methods

In terms of comparison between Supervised and Unsupervised accident detection methods, the studies of [15] showed the superiority of supervised methods, comprising Coarse-to-Fine clustering [15], Fully Connected (FC) [44], LSTM [82], Encoder-Decoder [83] and TRN [84] against those unsupervised counterparts such as ConvAE [49], ConvLSTM [85], AnoPred [36], AnoPred+Mask, FOL-AvgSTD+MarginLearning (FOL-AvgSTD+ML) and Ensemble (all in [44]), and FOL-AvgSTD [22]. Their research justified the effectiveness of methods which take advantage of motion features (optical flow) in comparison of those who just consider appearance features. Furthermore, they proved the superiority of their method due to combining supervised learning with motion features. Lastly, [15] is the only method which tries to provide a real-time VANET method by comparing the time and performance with other TAD methods. With all the strengths that have been mentioned in [15], still there is a lack of suggestion new feature extraction methods in their work.

D. Overall Comparison of State-of-the-Art Methods

By and Large, with regards to details of different methods, the concept of state-of-the-art self-supervised method [37] and unsupervised method [38] is similar as they extract appearance and motion features of objects and apply some further processing to reach their final features, which they later apply to train their accident detection models. By contrast, in supervised methods, while the approach of [29] is simplistic and cannot cover many actual accident scenarios, the VANET approach of [15] is considered more reliable. Despite all mentioned, what is missing in the state-of-the-art methods of TAD are

TABLE II
EVALUATION METRICS

Name	Description	Advantage	Limit	Applied in
Qualitative Analysis	Visual assessment	Gives an intuitive understanding of the scene	There is no unit for evaluation	[37], [22], [38], [29]
Area under ROC curve (AUC)	AUC of Temporal anomaly score of frames in a video	Determines temporal location of possible accident	Does not give spatial information within anomalous frame	[37], [22], [38], [15]
Accident Detection Rate $\frac{TP}{TP+FN}$	The ratio of detected accidents over all accidents	Shows the performance of model in VAD	Does not provide information about false detection	[29]
False Alarm Rate $\frac{FP}{TP+FN}$	The ratio of wrong accident alarms over all alarms	Shows the ratio of wrong VAD	Does not provide information about accident detection	[29]
Time Parameters	Number of Parameters, Training Time, and Average Detection Time	Shows the complexity and performance of the model	Does not give information about accuracy of model	[15]
Success rate curve	AUC of temporal overlap score of bounding boxes in a video	Determines the quality of object tracking in the model	Does not give spatial information within anomalous frame	[37]
Average Precision (AP)	Average between ratios of correctly identified accident frames to all detected accidents in all frames of a video for a specific class	Shows the performance of the model for one class or in Binary classification	Does not provide any information about false detection	[37], [38]
mean Average Precision (mAP)	Mean of AP's between all classes in videos	Shows the overall performance of the model across all classes	Does not provide information about false detection	[37], [38]
Anomaly-to-Accident (A2A)	A temporal window starting when accident participant appears in scene until collision	Can be used for accident anticipation	There may be a long interval between appearance time of accident participant and the time it start causing anomaly in the scene	[37]
Spatial-Temporal Area Under Curve (STAUC)	Area under the plot of spatial-temporal true positive rate (STTPR) and false positive rate	Determines the accuracy of models in detecting spatial-temporal location of anomaly	Requires dataset with corresponding annotation in time and location in each frame	[38]

influential features that can distinguish foreground accidental objects from dynamic background of the scene.

E. Introducing Baseline Methods in TAD Comparison Table

As AUC of ROC is the most accepted evaluation metric in the anomaly detection methods, and consequently in accident detection field, and while all the state-of-the-art did not apply their approach on a common dataset, the true comparison of them requires applying all to the same dataset. Having said that, contemplating the methods in general, supervised methods can achieve better results due to having the label of video frames (accident or no-accident) in comparison to unsupervised/self-supervised methods, though the unsupervised and self-supervised methods can better generalize their model for any case out of their available trained dataset. Therefore, to establish a robust reliable accident detection method, due to the limitation of available dataset and since it is unrealistic to have enough sample of each rare accident case, considering that there are 18 overall classes of accident based on DoTA dataset, the concentration of future research in this field would inevitably be in the unsupervised/self-supervised direction. Since accident detection can be considered as online binary action detection, labeled with normal and accident classes, according to [15], [38], and [44], some action detection methods were introduced as a supervised TAD. These methods include ResNet50 model [89] as their backbone to extract features and from there four different approaches

were applied, namely FC, LSTM, Encoder-Decoder, and TRN. The FC model is a three-layer Fully-Connected classification network, while LSTM is a one-layer LSTM to classify sequential images. As a modified version of LSTM, Encoder-Decoder is a model that classify current frames in the encoder part while in the decoder part future classes are predicted. As another version of Encoder-Decoder, Temporal recurrent network (TRN) [88] is having a feedback of future classes to the encoding part. In [37], another benchmark was used for comparison with their self-supervised method, namely DeepMask-FP-TAD[86]. This unsupervised approach utilizes a deep multi-branch mask network (DMMNet) applies fusion of features to effectively combine the benefits of optical flow registering and pixel-level image synthesizing methods. This results in a more flexible masking network for motion and appearance fusion in video frame prediction. Specifically, the mask layer in each branch is employed to adaptively adjust the magnitude range of estimated optical flow and the weight of predicted frames by optical flow registering and pixel-level image synthesizing, respectively. The comparison of TAD methods has been provided in the Table III.

VIII. FUTURE DIRECTIONS

Despite the mentioned advances, there are still some shortcomings in current methods. As explained in “Section VII - Discussion and Comparison”, still state-of-the-art methods are struggling with having a reliable object detection and

TABLE III
COMPARISON OF ACCIDENT DETECTION METHODS BASED ON TYPES, INPUTS, OBJECT DETECTION AND TRACKING,
AND EVALUATION METRICS ON DIFFERENT DATASET

Input	Method	Type	Object Detection	Object Tracking	Evaluations						
					SA	A3D	AUC DoTA	DADA-2000	A3D	mAP DADA-2000	STAUC DoTA
Gray	ConvAE [49]	Unsupervised	N/A	N/A	-	-	64.3	-	-	-	7.4
Flow	ConvLSTMAE [30]		N/A	N/A	50.4	49.5	66.3	-	-	-	7.9
Gray	AnoPred[36]		N/A	N/A	-	-	53.8	-	-	-	12.7
Flow	FOL-AvgIoU[22]		N/A	N/A	-	-	62.5	-	-	-	12.2
RGB	FOL-MinIoU[22]		Mask R-CNN	N/A	50.4	46.1	67.5	-	-	-	24.4
MaskedRGB	FOL-Mask[22]		Mask R-CNN	N/A	-	-	64.8	-	-	-	42.1
Box+Flow	FOL-AvgSTD[22]		Mask R-CNN	DeepSORT	53.4	49.7	+(Ego)61.2	-	-	-	+(Ego)34.6
	FOL-MaxSTD[22]		Mask R-CNN	DeepSORT	52.6	48.4	+(Ego)64.0	-	-	-	+(Ego)35.0
	FOL-AvgSTD+ML[44]		Mask R-CNN	DeepSORT	54.8	54.1	+(Ego)69.7	-	-	-	+(Ego)43.7
	FOL-MaxSTD[22]		Mask R-CNN	DeepSORT	55.8	59.3	+(Ego)69.7	-	-	-	+(Ego)43.7
Box+Flow	FOL-AvgSTD+ML[44]		Mask R-CNN	DeepSORT	55.6	60.1	+(Ego)69.7	-	-	-	+(Ego)43.7
RGB+Box+Flow+Ego	FOL-Ensemble (FOL-STD+AnoPred+Mask)[44]		Mask R-CNN	DeepSORT	-	-	69.7	-	-	-	43.7
RGB+Box+Flow	DeepMask-FP-TAD[86]		Mask R-CNN	DeepSORT	-	-	73.0	-	-	-	48.5
RGB+Box+Flow	SSC-TAD ^a [37]	Self-Supervised	Mask R-CNN	DeepSORT	-	59.0	-	67.5	59.9	40.4	-
RGB+Box+Flow	SSC-TAD ^{am} [37]		Mask R-CNN	DeepSORT	-	65.4	-	64.6	61.2	38.4	-
RGB+Box+Flow	SSC-TAD ^{amc} [37]		Mask R-CNN	DeepSORT	-	65.8	-	65.2	61.5	40.3	-
RGB	FC[44]	Supervised	N/A	N/A	-	67.8	-	66.5	64.1	42.7	-
RGB	LSTM[82]		N/A	N/A	-	61.7	-	-	-	9.9	-
RGB	Encoder-Decoder[87]		N/A	N/A	-	63.7	-	-	-	12.9	-
RGB	TRN[88]		N/A	N/A	-	73.6	-	-	-	14.5	-
Box	Movement Parameters method[29] ¹		Mask R-CNN	Centroid Tracking	-	78.0	-	-	-	15.3	-
RGB+Flow	Coarse-to-Fine clustering[15] ²		Faster R-CNN	N/A	-	79.3	-	-	-	-	-

¹ The custom dataset of [29] is not clarified and the ADR and FAR metrics were only used in this paper.

² Time evaluations were only provided in [15].

object tracking technique to build their TAD method upon them. Besides, spatio-temporal localization of anomalies in driving scenarios can be a helpful way of improving TAD. Besides, future of the field of accident detection lies within the unsupervised and self-supervised approaches. Such methods can be applied to extract the labels of videos directly from the video scenes and makes the model independent from any manual labeling of video frames, whether labeling the accidental/normal frame or determining the bounding boxes in each frame. Another prospect direction is TAD video generation, which requires powerful computational system. With such an approach, TAD becomes independent of data collection. Since occlusion has been one of the main drawbacks in detection of accidents, this issue should be investigated itself as a future direction. The real-time detection of accidents is one of the other aspect to prepare a TAD for real scenario of autonomous vehicles. For this purpose, models can be improved to extract features on possible accident scenarios, instead of continuously computation of features from objects in the scene. Moreover, as DoTA has classified different accidents, a possible direction is the fusion of models each of which is well adapted with limited number of accident categories and the merged model be able to deal with different scenarios by activating relevant sub-model. Furthermore, any TAD method that extracts reliable feature labels to detect accident can be tried to predict accident too. As another future direction, understanding different actions that cause accidents can result in a reliable accident detection/prediction too. All being said, literature of accident detection and prediction still lack a cross-modal dataset, in other words, there is not any dataset that provide accidents and other sources of information including GPS, LiDAR, radar and connected vehicles dataset that can help improve the performance of the models.

Additional to the mentioned TAD works and possible future directions, based on the provided information on tables of measuring units, available dataset and performance of TAD

methods, disunity of current state-of-the-art methods becomes clear with regards to different definition for annotation, creating different dataset and consequently, providing results on non-unanimous dataset. Should the challenge of accident detection and consequently accident anticipation are expected to solve, it is of utmost important to converge all research of TAD by introducing related protocols within intelligent transportation society. Such a general consensus may include but not limited to agreement on the accurate definition of accident, accident interval, before and after accident interval, which all together can determine the interval of each accident clip for dataset creation. What is more, as accidents are rare to happen, the field of TAD deeply suffer from a comprehensive dataset that could provide additional information rather than the video footage of accident scenes. Such an extra knowledge can be time, location (suburbs or downtown, highway or local road, traffic report of road ahead(congested, heavy traffic, light traffic, free-flowing) and etc.

IX. CONCLUSION

This study highlights the primary challenges in accident detection using dashcam videos. Firstly, because of rapid motion of the ego-car, distinguishing visually reconstructed current or future RGB frames from normal non-accidental scenes, that are given to model in training, is difficult. Secondly, all the TAD methods rely on object detection and object tracking as their fundamental backbone of their algorithms and those that were applied in TAD methods still down perform in case of occlusions and sudden appearance change of incidental objects in the scene. Thirdly, in order to apply TAD algorithms in real scenarios of autonomous vehicles, they should perform accurately in real time which has not yet addressed in literature except in the Coarse-to-Fine clustering method.

As video processing has a high computational cost, based on the categories provided in Figure 3, all state-of-the-art

methods tried to address TAD issue using frame-based techniques. Furthermore, based on the literature, it is clear that the problem of TAD with numerous types of accident should be pursued with unsupervised/self-supervised methods, as it is non-pragmatic to label all TAD videos and build a model that can generalize to unseen accident types. To address some of these challenges, Unsupervised Future Object Localization (FOL-TAD) was introduced a baseline for TAD where accident was detected by anomaly in accuracy and consistency of anticipated objects on future frames of the scene. SSC is considered as a complimented version of that baseline by suggesting a graph-based fusion of objects' and frame's features. To improve their initial baseline method, authors of FOL-TAD showed an improvement on their method by proposing the FOL-Ensemble method. Laying emphasis on time efficiency aspect, Coarse-to-Fine framework was introduced to address the real-time performance aspect of TAD.

Finally, this study provided all available dataset and evaluation metric in the field and introduced strengths and downsides of current methods and by suggesting future direction, depicted an overall road map for TAD.

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REFERENCES

- [1] *Road Accidents and Safety Statistics*, Government United Kingdom, London, U.K., 2023.
- [2] S. Singh, "Critical reasons for crashes investigated in the national motor vehicle crash causation survey," Nat. Center Statist. Anal., New Jersey, NJ, USA, Tech. Rep. DOT HS 812 506, 2015. [Online]. Available: <https://trid.trb.org/view/1507603>
- [3] *On-Road Automated Vehicle Standards Committee: Taxonomy and Definitions for Terms Related to On-Road Motor Vehicle Automated Driving Systems, Information Report*, SAE Int., Warrendale, PA, USA, 2014.
- [4] J. Guo, U. Kurup, and M. Shah, "Is it safe to drive? An overview of factors, metrics, and datasets for driveability assessment in autonomous driving," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 8, pp. 3135–3151, Aug. 2020.
- [5] S. Riedmaier, T. Ponn, D. Ludwig, B. Schick, and F. Diermeyer, "Survey on scenario-based safety assessment of automated vehicles," *IEEE Access*, vol. 8, pp. 87456–87477, 2020.
- [6] E. Yurtsever, J. Lambert, A. Carballo, and K. Takeda, "A survey of autonomous driving: Common practices and emerging technologies," *IEEE Access*, vol. 8, pp. 58443–58469, 2020.
- [7] D. Xiao, W. G. Geiger, H. Y. Yatbaz, M. Dianati, and R. Woodman, "Detecting hazardous events: A framework for automated vehicle safety systems," in *Proc. IEEE 25th Int. Conf. Intell. Transp. Syst. (ITSC)*, Oct. 2022, pp. 641–646.
- [8] F.-H. Chan, Y.-T. Chen, Y. Xiang, and M. Sun, "Anticipating accidents in dashcam videos," in *Proc. 13th Asian Conf. Comput. Vis.*, Taipei, Taiwan. Cham, Switzerland: Springer, 2017, pp. 136–153. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-319-54190-7_9#citeas
- [9] Y. Yao, M. Xu, C. Choi, D. J. Crandall, E. M. Atkins, and B. Dariush, "Egocentric vision-based future vehicle localization for intelligent driving assistance systems," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2019, pp. 9711–9717.
- [10] E. Yurtsever et al., "Risky action recognition in lane change video clips using deep spatiotemporal networks with segmentation mask transfer," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, Oct. 2019, pp. 3100–3107.
- [11] C. Li, S. H. Chan, and Y.-T. Chen, "Who make drivers stop? Towards driver-centric risk assessment: Risk object identification via causal inference," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2020, pp. 10711–10718.
- [12] G. Plumb, M. T. Ribeiro, and A. Talwalkar, "Finding and fixing spurious patterns with explanations," 2021, *arXiv:2106.02112*.
- [13] S. Stabinger, D. Peer, J. Piater, and A. Rodríguez-Sánchez, "Evaluating the progress of deep learning for visual relational concepts," *J. Vis.*, vol. 21, no. 11, p. 8, Oct. 2021.
- [14] G. Puebla and J. S. Bowers, "Can deep convolutional neural networks support relational reasoning in the same-different task?" *J. Vis.*, vol. 22, no. 10, p. 11, Sep. 2022.
- [15] Z. Zhou, X. Dong, Z. Li, K. Yu, C. Ding, and Y. Yang, "Spatio-temporal feature encoding for traffic accident detection in VANET environment," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 19772–19781, Oct. 2022.
- [16] S. S. Khan and M. G. Madden, "One-class classification: Taxonomy of study and review of techniques," *Knowl. Eng. Rev.*, vol. 29, no. 3, pp. 345–374, Jun. 2014.
- [17] L. Ruff et al., "Deep one-class classification," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 4393–4402.
- [18] P. Perera and V. M. Patel, "Learning deep features for one-class classification," *IEEE Trans. Image Process.*, vol. 28, no. 11, pp. 5450–5463, Nov. 2019.
- [19] D. M. J. Tax and R. P. W. Duin, "Support vector data description," *Mach. Learn.*, vol. 54, no. 1, pp. 45–66, Jan. 2004.
- [20] P. Zheng, S. Yuan, X. Wu, J. Li, and A. Lu, "One-class adversarial nets for fraud detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, 2019, pp. 1286–1293.
- [21] M. Sabokrou, M. Khalooei, M. Fathy, and E. Adeli, "Adversarially learned one-class classifier for novelty detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3379–3388.
- [22] Y. Yao, M. Xu, Y. Wang, D. J. Crandall, and E. M. Atkins, "Unsupervised traffic accident detection in first-person videos," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Nov. 2019, pp. 273–280.
- [23] Y. Zhu and S. Newsam, "Motion-aware feature for improved video anomaly detection," 2019, *arXiv:1907.10211*.
- [24] W. Sultani, C. Chen, and M. Shah, "Real-world anomaly detection in surveillance videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6479–6488.
- [25] S. Lin, H. Yang, X. Tang, T. Shi, and L. Chen, "Social MIL: Interaction-aware for crowd anomaly detection," in *Proc. 16th IEEE Int. Conf. Adv. Video Signal Based Surveill. (AVSS)*, Sep. 2019, pp. 1–8.
- [26] F. Landi, C. G. M. Snoek, and R. Cucchiara, "Anomaly locality in video surveillance," 2019, *arXiv:1901.10364*.
- [27] H. Zenati, C. Sheng Foo, B. Lecouat, G. Manek, and V. R. Chandrasekhar, "Efficient GAN-based anomaly detection," 2018, *arXiv:1802.06222*.
- [28] T. Schlegl, P. Seeböck, S. M. Waldstein, U. Schmidt-Erfurth, and G. Langs, "Unsupervised anomaly detection with generative adversarial networks to guide marker discovery," in *Proc. 25th Int. Conf. Inf. Process. Med. Imag.*, Boone, NC, USA. Cham, Switzerland: Springer, 2017, pp. 146–157. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-319-59050-9_12#citeas
- [29] D. Chand, S. Gupta, and I. Kavati, "Computer vision based accident detection for autonomous vehicles," in *Proc. IEEE 17th India Council Int. Conf. (INDICON)*, Dec. 2020, pp. 1–6.
- [30] Y. S. Chong and Y. H. Tay, "Abnormal event detection in videos using spatiotemporal autoencoder," in *Proc. 14th Int. Symp. Adv. Neural Netw.*, Hokkaido, Japan. Cham, Switzerland: Springer, 2017, pp. 189–196. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-319-59081-3_23#citeas
- [31] J. An and S. Cho, "Variational autoencoder based anomaly detection using reconstruction probability," *Special Lect. IE*, vol. 2, no. 1, pp. 1–18, Dec. 2015.
- [32] Y. Zhao, B. Deng, C. Shen, Y. Liu, H. Lu, and X.-S. Hua, "Spatio-temporal autoencoder for video anomaly detection," in *Proc. 25th ACM Int. Conf. Multimedia*, Oct. 2017, pp. 1933–1941.
- [33] D. Gong et al., "Memorizing normality to detect anomaly: Memory-augmented deep autoencoder for unsupervised anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1705–1714.
- [34] I. Goodfellow, "NIPS 2016 tutorial: Generative adversarial networks," 2017, *arXiv:1701.00160*.
- [35] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Comput. Surv.*, vol. 41, no. 3, pp. 1–58, Jul. 2009.
- [36] W. Liu, W. Luo, D. Lian, and S. Gao, "Future frame prediction for anomaly detection—A new baseline," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6536–6545.
- [37] J. Fang, J. Qiao, J. Bai, H. Yu, and J. Xue, "Traffic accident detection via self-supervised consistency learning in driving scenarios," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 9601–9614, Jul. 2022.

- [38] Y. Yao et al., "DoTA: Unsupervised detection of traffic anomaly in driving videos," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 444–459, Jan. 2023.
- [39] D. Xiao, M. Dianati, W. G. Geiger, and R. Woodman, "Review of graph-based hazardous event detection methods for autonomous driving systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 5, pp. 4697–4715, May 2023.
- [40] J. Qiao, J. Fang, D. Yan, and J. Xue, "Driving accident detection by self-supervised adversarial appearance-motion prediction in first-person videos," in *Proc. 3rd Int. Conf. Unmanned Syst. (ICUS)*, Nov. 2020, pp. 1083–1088.
- [41] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3645–3649.
- [42] A. Dosovitskiy et al., "FlowNet: Learning optical flow with convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 2758–2766.
- [43] M. Mathieu, C. Couprie, and Y. LeCun, "Deep multi-scale video prediction beyond mean square error," 2015, *arXiv:1511.05440*.
- [44] Y. Yao, X. Wang, M. Xu, Z. Pu, E. Atkins, and D. Crandall, "When, where, and what? A new dataset for anomaly detection in driving videos," 2020, *arXiv:2004.03044*.
- [45] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2016, pp. 3464–3468.
- [46] R. Herzog et al., "Spatio-temporal action graph networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 2347–2356.
- [47] J. Fang, D. Yan, J. Qiao, J. Xue, and H. Yu, "DADA: Driver attention prediction in driving accident scenarios," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 4959–4971, Jun. 2022.
- [48] C. Lu, J. Shi, and J. Jia, "Abnormal event detection at 150 FPS in MATLAB," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2720–2727.
- [49] M. Hasan, J. Choi, J. Neumann, A. K. Roy-Chowdhury, and L. S. Davis, "Learning temporal regularity in video sequences," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 733–742.
- [50] W. Luo, W. Liu, and S. Gao, "A revisit of sparse coding based anomaly detection in stacked RNN framework," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 341–349.
- [51] R. Morais, V. Le, T. Tran, B. Saha, M. Mansour, and S. Venkatesh, "Learning regularity in skeleton trajectories for anomaly detection in videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11996–12004.
- [52] R. T. Ionescu, F. S. Khan, M.-I. Georgescu, and L. Shao, "Object-centric auto-encoders and dummy anomalies for abnormal event detection in video," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7842–7851.
- [53] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow Twins: Self-supervised learning via redundancy reduction," in *Proc. 38th Int. Conf. Mach. Learn.*, vol. 139, M. Meila and T. Zhang, Eds., Jul. 2021, pp. 12310–12320.
- [54] S. Haresh, S. Kumar, M. Z. Zia, and Q.-H. Tran, "Towards anomaly detection in dashcam videos," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Oct. 2020, pp. 1407–1414.
- [55] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.
- [56] W. Li, V. Mahadevan, and N. Vasconcelos, "Anomaly detection and localization in crowded scenes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 1, pp. 18–32, Jan. 2014.
- [57] X. Wang and A. Gupta, "Videos as space-time region graphs," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 399–417.
- [58] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 1–9.
- [59] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.
- [60] E. Ilg, N. Mayer, T. Saikia, M. Keuper, A. Dosovitskiy, and T. Brox, "FlowNet 2.0: Evolution of optical flow estimation with deep networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2462–2470.
- [61] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.
- [62] R. W. Schafer, "What is a Savitzky–Golay filter? [Lecture notes]," *IEEE Signal Process. Mag.*, vol. 28, no. 4, pp. 111–117, Jul. 2011.
- [63] J. C. Nascimento, A. J. Abrantes, and J. S. Marques, "An algorithm for centroid-based tracking of moving objects," in *Proc. IEEE Int. Conf. Acoust. Speech, Signal Processing*, vol. 6, Mar. 1999, pp. 3305–3308.
- [64] P. Aggarwal, "Detecting lanes with OpenCV and testing on Indian roads," Medium, San Francisco, CA, USA, 2016. [Online]. Available: <https://medium.com/computer-car/my-lane-detection-project-for-the-self-driving-car-nanodegree-by-udacity-36a230553bd3>
- [65] B. D. Lucas and T. Kanade, "An iterative image registration technique with an application to stereo vision," in *Proc. 7th Int. Joint Conf. Artif. Intell.*, vol. 2, 1981, pp. 674–679.
- [66] W. Zhang, Q. M. J. Wu, Y. Yang, T. Akilan, and H. Zhang, "A width-growth model with subnetwork nodes and refinement structure for representation learning and image classification," *IEEE Trans. Ind. Informat.*, vol. 17, no. 3, pp. 1562–1572, Mar. 2021.
- [67] Y. Yang and Q. M. J. Wu, "Features combined from hundreds of midlayers: Hierarchical networks with subnetwork nodes," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 11, pp. 3313–3325, Nov. 2019.
- [68] L. Wei, X. Wang, A. Wu, R. Zhou, and C. Zhu, "Robust subspace segmentation by self-representation constrained low-rank representation," *Neural Process. Lett.*, vol. 48, no. 3, pp. 1671–1691, Dec. 2018.
- [69] J. Shi and J. Malik, "Normalized cuts and image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 22, no. 8, pp. 888–905, Aug. 2000.
- [70] W. Bao, Q. Yu, and Y. Kong, "Uncertainty-based traffic accident anticipation with spatio-temporal relational learning," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 2682–2690.
- [71] J. Fang, D. Yan, J. Qiao, J. Xue, H. Wang, and S. Li, "DADA-2000: Can driving accident be predicted by driver attention? Analyzed by a benchmark," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, Oct. 2019, pp. 4303–4309.
- [72] F. Yu et al., "BDD100K: A diverse driving dataset for heterogeneous multitask learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2636–2645.
- [73] H. Pradana, M.-S. Dao, and K. Zettsu, "Augmenting ego-vehicle for traffic near-miss and accident classification dataset using manipulating conditional style translation," 2023, *arXiv:2301.02726*.
- [74] W. G. Najm, J. D. Smith, and M. Yanagisawa, "Pre-crash scenario typology for crash avoidance research," U.S. Dept. Transp., Nat. Highway Traffic Saf. Admin., Washington, DC, USA, Tech. Rep. DOT HS 810 767, 2007.
- [75] T. You and B. Han, "Traffic accident benchmark for causality recognition," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Cham, Switzerland: Springer, 2020, pp. 540–556. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-030-58571-6_32#citeas
- [76] H. Kataoka, T. Suzuki, S. Oikawa, Y. Matsui, and Y. Satoh, "Drive video analysis for the detection of traffic near-miss incidents," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2018, pp. 3421–3428.
- [77] W. Luo, W. Liu, and S. Gao, "Remembering history with convolutional LSTM for anomaly detection," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2017, pp. 439–444.
- [78] V. Mahadevan, W. Li, V. Bhalodia, and N. Vasconcelos, "Anomaly detection in crowded scenes," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 1975–1981.
- [79] Y. Wu, J. Lim, and M.-H. Yang, "Online object tracking: A benchmark," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 2411–2418.
- [80] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.
- [81] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, "MVTec AD—A comprehensive real-world dataset for unsupervised anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9592–9600.
- [82] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
- [83] C. Zhang et al., "A deep neural network for unsupervised anomaly detection and diagnosis in multivariate time series data," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, 2019, pp. 1409–1416.
- [84] M. Canizo, I. Triguero, A. Conde, and E. Onieva, "Multi-head CNN-RNN for multi-time series anomaly detection: An industrial case study," *Neurocomputing*, vol. 363, pp. 246–260, Oct. 2019.
- [85] J. R. Medel and A. Savakis, "Anomaly detection in video using predictive convolutional long short-term memory networks," 2016, *arXiv:1612.00390*.
- [86] S. Li, J. Fang, H. Xu, and J. Xue, "Video frame prediction by deep multi-branch mask network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 4, pp. 1283–1295, Apr. 2021.

- [87] K. Cho et al., "Learning phrase representations using RNN encoder-decoder for statistical machine translation," 2014, *arXiv:1406.1078*.
- [88] M. Xu, M. Gao, Y.-T. Chen, L. Davis, and D. Crandall, "Temporal recurrent networks for online action detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5532–5541.
- [89] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.



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